A Lightweight RL-Driven Deep Unfolding Network for Robust WMMSE Precoding

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***Abstract*— Weighted Minimum Mean Square Error (WMMSE) precoding can achieve near-optimal weighted sum rate (WSR) in MU-MIMO-OFDM systems, but it suffers from high computational complexity and severe dependence on accurate channel state information (CSI). This letter proposes a lightweight reinforcement learning (RL)-driven deep unfolding (DU) network (RLDDU-Net) to overcome these limitations. Specifically, its DU module maps a wideband stochastic WMMSE algorithm, which maximizes the ergodic WSR under imperfect CSI, into a deep-learning framework. Approximation techniques exploiting beam-domain sparsity and subcarrier correlation are also adopted to significantly reduce complexity and accelerate convergence. The RL module dynamically generates compensation matrices to mitigate approximation errors and adjusts DU network depth online, thereby enhancing both flexibility and performance. Simulation results demonstrate that RLDDU-Net outperforms existing baselines in terms of ergodic WSR, computational efficiency, and convergence speed under imperfect CSI.**

***Index Terms*— WMMSE, imperfect CSI, deep-unfolding, RL.**

I. Introduction

THE 6G wireless communication adopts massive multi-user (MU) multiple-input multiple-output (MIMO) and orthogonal frequency-division multiplexing (OFDM) technologies to support rapid data growth. Weighted Minimum Mean Square Error (WMMSE) precoding has been widely recognized for enabling near-optimal weighted sum-rate performance [1]. However, the practical deployment of WMMSE in massive MU-MIMO-OFDM faces two critical challenges [2]. First, it heavily relies on accurate channel state information (CSI), which is often unavailable due to limited pilot resources and channel aging. Second, its high computational complexity must be reduced to meet latency and deployment constraints.

To address the first challenge, [3], [4], [5] modeled imperfect CSI using Gaussian-distributed posterior channel models and formulated the robust precoding problem as an ergodic weighted sum-rate (EWSR) maximization problem under a sum power constraint (SPC). Specifically, [3] and [4] developed stochastic WMMSE (SWMMSE) based on the sample average approximation (SAA) method, which alternately optimizes the receiver, weighting, and precoding matrices using virtual CSI samples drawn from channel error statistics. Reference [5] iteratively computed deterministic equivalents of the EWSR and optimized the precoders via the majorization-minimization (MM) method. However, these methods either incur high per-iteration complexity scaling cubically with the number of transmit antennas or suffer from slow convergence due to nested iterations. To address the second challenge, some robust linear precoding schemes, such as regularized zero-forcing (RZF) [6] and signal-to-leakage-and-noise ratio (SLNR) precoding [7], have been proposed but generally deliver sub-optimal performance compared to WMMSE-based methods because they do not directly optimize the EWSR objective [5]. Data-driven approaches such as [8] and [9] employed multi-layer perceptrons (MLPs) and convolutional neural networks (CNNs) to approximate the WMMSE algorithm, but suffer from limited interpretability and generalization. More recently, [2] proposed a practical WMMSE network (PO-WMMSE Net) that combines deep unfolding and approximation techniques to reduce complexity and accelerate convergence. By leveraging domain knowledge and introducing only a small number of trainable compensation matrices, it achieves stable performance across diverse scenarios. Nevertheless, it employs fixed compensation matrices and a uniform network depth for all channel conditions, limiting its flexibility under highly dynamic CSI. Moreover, all the aforementioned methods are designed for narrowband systems. When extended to wideband systems, they must be applied independently to each subcarrier, resulting in prohibitive computational overhead.

To address these challenges, this letter first develops a wideband variant of the SWMMSE algorithm [3] and then enhances it into a practical-oriented, lightweight reinforcement learning-driven deep-unfolding network (RLDDU-Net). Specifically, RLDDU-Net consists of a deep-unfolding (DU) module and a reinforcement learning (RL) module. The DU module maps each iteration of the wideband robust WMMSE into a network layer and applies approximation techniques to significantly reduce computational complexity by leveraging beam-domain sparsity and subcarrier correlation. These approximations also enable the direct use of CSI statistics, avoiding the sample average approximation (SAA) and nested iterations required in conventional robust WMMSE methods, thereby accelerating convergence. The RL module adaptively generates compensation matrices to mitigate approximation errors and adjusts the network depth in response to varying CSI conditions. Numerical results under imperfect CSI demonstrate that the proposed RLDDU-Net significantly outperforms baseline approaches.

II. System Model and Problem Formulation

As illustrated in Fig. 1, we consider a massive MU-MIMO-OFDM system operating in time-division duplexing (TDD)

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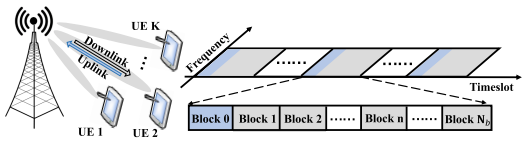

Fig. 1. System model and time slot diagram.

mode, where a base station (BS) with a uniform planar array (UPA) of M_t-antennas serves K users, each with M_r-antennas ($M_r \ll M_t$). Specifically, the time domain is divided into multiple timeslots, each further split into (N_b+1) blocks: the 0-th block is used for uplink training, and the remaining N_b blocks are reserved for downlink transmission. In the frequency domain, every N_f adjacent subcarriers (typically $N_f = 48$ in 5G [10]) form a resource block group (RBG), which serves as the granularity for wideband precoding. Let $\mathcal{K} \triangleq \{1,2,\ldots,K\}$ and $\mathcal{F} \triangleq \{1,2,\ldots,N_f\}$ denote the sets of user and subcarrier indices, respectively. In the following, we consider one timeslot and one RGB, for example.

Let $\mathbf{H}^{\mathbf{b}}_{k,f,n} \in \mathbb{C}^{M_r\times M_t}$ denote the beam-domain channel from the BS to user k at block n and subcarrier f. The corresponding antenna-domain channel is $\mathbf{H}_{k,f,n} = \mathbf{H}^{\mathbf{b}}_{k,f,n}\mathbf{\Phi}$, where $\mathbf{\Phi} \in \mathbb{C}^{M_t\times M_t}$ is the array response matrix, typically modeled as a normalized discrete Fourier transform (DFT) matrix satisfying $\mathbf{\Phi}\mathbf{\Phi}^H = \mathbf{I}$ for large-scale uniform planar. The channel is assumed to follow flat block fading per subcarrier. During the uplink training block ($n=0$), advanced beam-domain channel estimation techniques can be applied at the BS to obtain an estimate $\bar{\mathbf{H}}^{\mathbf{b}}_{k,f,0}$ of $\mathbf{H}^{\mathbf{b}}_{k,f,0}$ with low pilot overhead [11]. In downlink blocks ($n=1,\ldots,N_b$), channel aging caused by user mobility and other dynamics can be modeled using a first-order Gauss-Markov process [5]:

$$\mathbf{H}^{\mathbf{b}}_{k,f,n} = \beta_{k,f,n}\mathbf{H}^{\mathbf{b}}_{k,f,0} + \sqrt{1-\beta^2_{k,f,n}}\mathbf{M}_{k,f}\odot\mathbf{W}_{k,f,n}, \quad (1)$$

where $\odot$ is the element-wise multiplication, $\mathbf{M}_{k,f} \in \mathbb{C}^{M_r\times M_t}$ denotes a non-negative amplitude profile matrix, and $\mathbf{W}_{k,f,n} \in \mathbb{C}^{M_r\times M_t}$ is an i.i.d complex Gaussian noise matrix with zero mean and unit variance entries. The temporal correlation coefficient $\beta_{k,f,n} \in (0,1)$ and $\mathbf{\Omega}_{k,f} \triangleq \mathbf{M}_{k,f}\odot\mathbf{M}_{k,f}$ can be acquired through the sounding process. Accordingly, the BS can infer the posterior distribution of $\mathbf{H}^{\mathbf{b}}_{k,f,n}$ as:

$$\mathbf{H}^{\mathbf{b}}_{k,f,n} \sim \mathcal{CN}\big(\bar{\mathbf{H}}^{\mathbf{b}}_{k,f,n}, \Delta\mathbf{H}^{\mathbf{b}}_{k,f,n}\big), \quad (2)$$

where $\bar{\mathbf{H}}^{\mathbf{b}}_{k,f,n} = \beta_{k,f,n}\bar{\mathbf{H}}^{\mathbf{b}}_{k,f,0}$ and $\Delta\mathbf{H}^{\mathbf{b}}_{k,f,n} = (1-\beta^2_{k,f,n})\mathbf{\Omega}_{k,f}$. Note that a larger n indicates a longer time interval since the last uplink training, resulting in more severe channel aging (i.e., smaller $\beta_{k,f,n}$ [5]) and less accurate CSI.

Based on $\{\bar{\mathbf{H}}^{\mathbf{b}}_{k,f}, \Delta\mathbf{H}^{\mathbf{b}}_{k,f}, \forall k,f\}$, the BS designs the wideband precoders $\{\mathbf{V}_k, \forall k\}$ for each downlink block, where $\mathbf{V}_k \in \mathbb{C}^{M_t\times M_r}$ is used to process the transmit signal $\boldsymbol{s}_{k,f} \in \mathbb{C}^{M_r\times 1}$ for user k and $\mathbb{E}[\boldsymbol{s}_{k,f}\boldsymbol{s}^H_{k,f}] = \mathbf{I}$. For simplicity, the index n is omitted when no ambiguity arises. Then, the received signal at user k and subcarrier f is given by:

$$\boldsymbol{y}_{k,f} = \mathbf{H}_{k,f}\mathbf{V}_k\boldsymbol{s}_{k,f} + \sum_{m\in\mathcal{K}/k}\mathbf{H}_{k,f}\mathbf{V}_m\boldsymbol{s}_{m,f} + \boldsymbol{n}_{k,f}, \quad (3)$$

where $\boldsymbol{n}_{k,f} \sim \mathcal{CN}(\mathbf{0}, \sigma^2_k\mathbf{I})$ denotes additive noise. The downlink transmission rate for user k at subcarrier f is given by

$$\mathcal{R}_{k,f} = \log\det\Big(\mathbf{I} + \mathbf{H}_{k,f}\mathbf{V}_k\mathbf{V}^H_k\mathbf{H}^H_{k,f} \cdot \big(\sum_{m\in\mathcal{K}/k}\mathbf{H}_{k,f}\mathbf{V}_m\mathbf{V}^H_m\mathbf{H}^H_{k,f} + \sigma^2_k\mathbf{I}\big)^{-1}\Big). \quad (4)$$

Finally, we formulate the wideband robust precoding problem as expected weighted sum-rate (EWSR) maximization, subject to a sum power constraint (SPC), following [3]:

$$\mathbf{P}_1 : \max_{\mathbf{V}} \sum_{k\in\mathcal{K}}\sum_{f\in\mathcal{F}}\omega_k\mathbb{E}[\mathcal{R}_{k,f}], \quad \text{s.t.} \sum_{k\in\mathcal{K}}\mathrm{Tr}(\mathbf{V}_k\mathbf{V}^H_k) \le P_{max}.$$

where ω_k represents the priority of user k, and $\mathbb{E}$ is the expectation taken over channel distributions (2).

III. The Algorithm and Network Design

A. Wideband Variant of SWMMSE Algorithm

We first develops a wideband variant of the SWMMSE in [3] to solve $\mathbf{P}_1$. To avoid directly handling the constraint, we incorporate it into the objective and consider problem $\mathbf{P}_2$:

$$\mathbf{P}_2 : \max_{\mathbf{V}} \sum_{k\in\mathcal{K}}\sum_{f\in\mathcal{F}}\omega_k\mathbb{E}[\tilde{\mathcal{R}}_{k,f}],$$

where

$$\tilde{\mathcal{R}}_{k,f} = \log\det\Big(\mathbf{I} + \mathbf{H}_{k,f}\mathbf{V}_k\mathbf{V}^H_k\mathbf{H}^H_{k,f} \big(\sum_{m\in\mathcal{K}/k}\mathbf{H}_{k,f}\mathbf{V}_m\mathbf{V}^H_m\mathbf{H}^H_{k,f} + \frac{\sigma^2_k}{P_{max}}\sum_{m\in\mathcal{K}}\mathrm{Tr}(\mathbf{V}_m\mathbf{V}^H_m)\mathbf{I}\big)^{-1}\Big). \quad (5)$$

According to the full power property [1, Proposition 3], any stationary point $\mathbf{V}^{**}_k$ of $\mathbf{P}_1$ corresponds to a stationary point $\mathbf{V}^*_k$ of $\mathbf{P}_2$ and satisfies $\mathbf{V}^{**}_k = \xi\mathbf{V}^*_k$ with $\xi = \sqrt{P_{max}}\big(\sum_{k=1}^K\mathrm{Tr}(\mathbf{V}^*_k(\mathbf{V}^*_k)^H)\big)^{-\frac{1}{2}}$. Therefore, $\mathbf{P}_1$ can be solved by addressing $\mathbf{P}_2$, followed by a scaling operation.

Then, $\mathbf{P}_2$ can be equivalently reformulated as the following weighted MMSE problem by introducing auxiliary variables $\{\mathbf{U}_{k,f}, \mathbf{W}_{k,f}\}$ according to [3, Lemma 1]:

$$\min_{\mathbf{V}}\mathbb{E}\Big[\min_{\{\mathbf{U},\mathbf{W}\}}\sum_{k\in\mathcal{K}}\sum_{f\in\mathcal{F}}\omega_k\Big(\mathrm{Tr}(\mathbf{W}_{k,f}\tilde{\mathbf{E}}_{k,f}) - \log\det\mathbf{W}_{k,f}\Big)\Big],$$

where $\tilde{\mathbf{E}}_{k,f} \triangleq (\mathbf{I} - \mathbf{U}^H_{k,f}\mathbf{H}_{k,f}\mathbf{V}_k)(\mathbf{I} - \mathbf{U}^H_{k,f}\mathbf{H}_{k,f}\mathbf{V}_k)^H + \sum_{m\in\mathcal{K}/\{k\}}\mathbf{U}^H_{k,f}\mathbf{H}_{k,f}\mathbf{V}_m\mathbf{V}^H_m\mathbf{H}^H_{k,f}\mathbf{U}_{k,f} + \sigma^2_k\mathbf{U}^H_{k,f}\mathbf{U}_{k,f}$. Noting that fixing any two of $\mathbf{U}, \mathbf{W}$, and $\mathbf{V}$ transforms $\mathbf{P}_3$ into a convex problem with a closed-form solution. We solve $\mathbf{P}_3$ using the block coordinate descent (BCD) [1] and SAA [3]. Specifically, at the i-th iteration, we sample a virtual channel realization $\mathbf{H}^i_{k,f}$ from the posterior distribution of $\mathbf{H}_{k,f}, \forall k,f$. Then, given $\{\mathbf{V}^{i-1}, \mathbf{H}^i\}$, the problem $\mathbf{P}_3$ with respect to variable $\mathbf{U}^i$ has the following closed-form solution:

$$\mathbf{U}^i_{k,f} = (\mathbf{A}^i_{k,f})^{-1}\mathbf{H}^i_{k,f}\mathbf{V}^{i-1}_k, \forall k,f, \quad (6)$$

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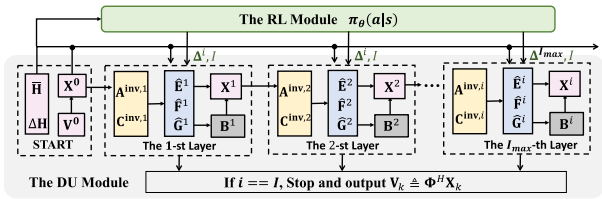

Fig. 2. The RLDDU-Net architecture.

where the variable $\mathbf{A}^i_{k,f} = \sum_{m=1}^K \mathbf{H}^i_{k,f}\mathbf{V}^{i-1}_m\big(\mathbf{H}^i_{k,f}\mathbf{V}^{i-1}_m\big)^H + \frac{\sigma_k^2}{P_{max}}\sum_{m=1}^K \mathrm{Tr}\big(\mathbf{V}^{i-1}_m\big(\mathbf{V}^{i-1}_m\big)^H\big)\mathbf{I}$. Once $\mathbf{U}^i$ is further fixed, the optimal $\mathbf{W}^i$ is derived as

$$\mathbf{W}^i_{k,f} = \big(\mathbf{I} - \big(\mathbf{U}^i_{k,f}\big)^H\mathbf{H}^i_{k,f}\mathbf{V}^{i-1}_k\big)^{-1}, \forall k, f. \tag{7}$$

Finally, setting the derivative of the objective function of $\mathbf{P}_2$ with respect to $\mathbf{V}_k$ to zero yields the optimal precoder:

$$\mathbf{V}^i_k = \big(\mathbf{B}^i\big)^{-1}\sum_{f\in\mathcal{F}}\mathbb{E}\big[\mathbf{H}^H_{k,f}\mathbf{U}^i_{k,f}\mathbf{W}^i_{k,f}\big], \forall k, \tag{8}$$

where $\mathbf{B}^i = \sum_{f=1}^{N_f}\sum_{m=1}^K \mathbb{E}\big[\frac{\sigma_m^2}{P_{max}}\mathrm{Tr}\big(\omega_m\mathbf{U}^i_{m,f}\mathbf{W}^i_{m,f}\mathbf{U}^{i\,H}_{m,f}\big)\mathbf{I} + \omega_m\mathbf{H}^H_{m,f}\mathbf{U}^i_{m,f}\mathbf{W}^i_{m,f}\mathbf{U}^{i\,H}_{m,f}\mathbf{H}_{m,f}\big]$. Although SAA is used, convergence of this algorithm to a stationary solution can be established following similar analysis in [3], owing to the average-over-iterations effect in stochastic optimization.

B. Lightweight RL-Driven Deep-Unfolding Network

Building on the wideband SWMMSE, we propose the lightweight RLDDU-Net, comprising a DU module and an RL module, as shown in Fig. 2. The DU module unfolds the wideband SWMMSE into a deep network, reducing computational overhead and accelerating convergence by exploiting subcarrier correlation and beam-domain sparsity. The RL module trains a policy $\pi_{\boldsymbol{\theta}}$ to adaptively generate compensation matrices and the unfolding depth, enhancing robustness under varying CSI accuracy levels.

1) The DU Module: To facilitate the exploitation of beam-domain sparsity, the DU module focuses on optimizing the beam-domain precoders $\big\{\mathbf{X}_k \triangleq \mathbf{\Phi}\mathbf{V}_k, \forall k\big\}$ instead of $\big\{\mathbf{V}_k, \forall k\big\}$. By substituting (6) and (7) into (8) and applying the matrix inversion lemma [12], we obtain that

$$\mathbf{X}^i_k = \big(\tilde{\mathbf{B}}^i\big)^{-1}\sum_{f\in\mathcal{F}}\mathbf{E}^i_{k,f}, \tag{9}$$

where $\tilde{\mathbf{B}}^i = \sum_{f=1}^{N_f}\sum_{m=1}^K \frac{\sigma_m^2}{P_{max}}\mathrm{Tr}\big(\mathbf{F}^i_{m,f}\big)\mathbf{I} + \mathbf{G}^i_{m,f}$. By defining $\mathbf{D}^i_{k,f} = \mathbf{H}^{\mathbf{b}}_{k,f}\mathbf{X}^{i-1}_k\big(\mathbf{X}^{i-1}_k\big)^H\big(\mathbf{H}^{\mathbf{b}}_{k,f}\big)^H$ and $\mathbf{C}^i_{k,f} = \mathbf{A}^i_{k,f} - \mathbf{D}^i_{k,f}$, the terms $\big\{\mathbf{E}^i_{k,f}, \mathbf{F}^i_{k,f}, \mathbf{G}^i_{k,f}\big\}$ are given by

$$\mathbf{E}^i_{k,f} = \mathbb{E}\big[\big(\mathbf{H}^{\mathbf{b}}_{k,f}\big)^H\big(\mathbf{C}^i_{k,f}\big)^{-1}\mathbf{H}^{\mathbf{b}}_{k,f}\big]\mathbf{X}^{i-1}_k, \tag{10}$$

$$\mathbf{F}^i_{k,f} = \mathbb{E}\big[\big(\mathbf{C}^i_{k,f}\big)^{-1}\mathbf{D}^i_{k,f}\big(\mathbf{A}^i_{k,f}\big)^{-1}\big], \tag{11}$$

$$\mathbf{G}^i_{k,f} = \mathbb{E}\big[\big(\mathbf{H}^{\mathbf{b}}_{k,f}\big)^H\big(\mathbf{C}^i_{k,f}\big)^{-1}\mathbf{D}^i_{k,f}\big(\mathbf{A}^i_{k,f}\big)^{-1}\mathbf{H}^{\mathbf{b}}_{k,f}\big]. \tag{12}$$

Note that the precoders $\big\{\mathbf{X}_k, \forall k\big\}$ are fully determined by expectation terms $\big\{\mathbf{E}^i_{k,f}, \mathbf{F}^i_{k,f}, \mathbf{G}^i_{k,f}, \forall k, f\big\}$. To alleviate the computational overhead involving a large number of subcarriers, we uniformly sample a subset $\tilde{\mathcal{F}} \in \mathcal{F}$ and compute $\big\{\mathbf{E}^i_{k,f}, \mathbf{F}^i_{k,f}, \mathbf{G}^i_{k,f}, \forall k\big\}$ only for $f \in \tilde{\mathcal{F}}$. For the remaining $f \in \mathcal{F}\backslash\tilde{\mathcal{F}}$, considering the strong correlation between adjacent subcarriers, we use second-order Lagrange interpolation to estimate the corresponding matrices. Specifically, given that $\mathbf{E}^i_{k,f_0}$, $\mathbf{E}^i_{k,f_1}$, and $\mathbf{E}^i_{k,f_2}$ are known at three subcarrier $f_0 < f_1 < f_2$, the value of $\mathbf{E}^i_{k,f}$ at any intermediate subcarrier $f \in \big(f_0, f_2\big)$ can be approximated as:

$$\mathbf{E}^i_{k,f} = l_0\mathbf{E}^i_{k,f_0} + l_1\mathbf{E}^i_{k,f_1} + l_2\mathbf{E}^i_{k,f_2}, \tag{13}$$

where the coefficients are defined as $l_0 = \frac{(f-f_1)(f-f_2)}{(f_0-f_1)(f_0-f_2)}$, $l_1 = \frac{(f-f_0)(f-f_2)}{(f_1-f_0)(f_1-f_2)}$, and $l_2 = \frac{(f-f_0)(f-f_1)}{(f_2-f_0)(f_2-f_1)}$. The same interpolation method also applies to $\mathbf{F}^i_{k,f}$ and $\mathbf{G}^i_{k,f}$.

In the following, we further reduce the computational complexity of processing the sampled subset of subcarriers. First, considering that $\mathbf{A}^i_{k,f}$ and $\mathbf{C}^i_{k,f}$ tend to be diagonal when $M_t \gg M_r$, we approximate $\mathbb{E}\big[\big(\mathbf{A}^i_{k,f}\big)^{-1}\big]$ and $\mathbb{E}\big[\big(\mathbf{C}^i_{k,f}\big)^{-1}\big]$ by $\mathbf{A}^{\mathbf{inv},i}_{k,f} + \mathbf{Z}^{\mathbf{A},i}_{k,f}$ and $\mathbf{C}^{\mathbf{inv},i}_{k,f} + \mathbf{Z}^{\mathbf{C},i}_{k,f}$, respectively. Here, $\mathbf{A}^{\mathbf{inv},i}_{k,f}$ and $\mathbf{C}^{\mathbf{inv},i}_{k,f}$ are first-order diagonal Taylor approximations for $\mathbb{E}\big[\big(\mathbf{A}^i_{k,f}\big)^{-1}\big]$ and $\mathbb{E}\big[\big(\mathbf{C}^i_{k,f}\big)^{-1}\big]$, given by:

$$\mathbf{A}^{\mathbf{inv},i}_{k,f} = 2\mathbb{E}\big[\mathbf{A}^i_{k,f}\big]^+ - \mathbb{E}\big[\mathbf{A}^i_{k,f}\big]^+\mathbb{E}\big[\mathbf{A}^i_{k,f}\big]\mathbb{E}\big[\mathbf{A}^i_{k,f}\big]^+, \tag{14}$$

$$\mathbf{C}^{\mathbf{inv},i}_{k,f} = 2\mathbb{E}\big[\mathbf{C}^i_{k,f}\big]^+ - \mathbb{E}\big[\mathbf{C}^i_{k,f}\big]^+\mathbb{E}\big[\mathbf{C}^i_{k,f}\big]\mathbb{E}\big[\mathbf{C}^i_{k,f}\big]^+, \tag{15}$$

where $\mathbf{Y}^+$ denotes the element-wise reciprocal of $\mathbf{Y}$' diagonal entries with off-diagonal elements set to zero, and compensation matrices $\big\{\mathbf{Z}^{\mathbf{A},i}_{k,f}, \mathbf{Z}^{\mathbf{C},i}_{k,f}\big\}$ are introduced to narrow the gap between $\big\{\mathbf{A}^{\mathbf{inv},i}_{k,f}, \mathbf{C}^{\mathbf{inv},i}_{k,f}\big\}$ and $\big\{\mathbb{E}\big[\big(\mathbf{A}^i_{k,f}\big)^{-1}\big], \mathbb{E}\big[\big(\mathbf{C}^i_{k,f}\big)^{-1}\big]\big\}$. Then, using compensation matrices $\big\{\mathbf{O}^{\mathbf{E},i}_{k,f}, \mathbf{O}^{\mathbf{F},i}_{k,f}, \mathbf{O}^{\mathbf{E},i}_{k,f}\big\}$ to replace trivial cross-related terms, the expectation terms (10)-(12) are approximated as:

$$\hat{\mathbf{E}}^i_{k,f} = \mathbb{E}\big[\big(\mathbf{H}^{\mathbf{b}}_{k,f}\big)^H\big(\mathbf{C}^{\mathbf{inv},i}_{k,f} + \mathbf{O}^{\mathbf{E},i}_{k,f}\big)\mathbf{H}^{\mathbf{b}}_{k,f}\big]\mathbf{X}^{i-1}_k, \tag{16}$$

$$\hat{\mathbf{F}}^i_{k,f} = \mathbf{C}^{\mathbf{inv},i}_{k,f}\mathbb{E}\big[\mathbf{D}^i_{k,f}\big]\mathbf{A}^{\mathbf{inv},i}_{k,f} + \mathbf{O}^{\mathbf{F},i}_{k,f}, \tag{17}$$

$$\hat{\mathbf{G}}^i_{k,f} = \mathbb{E}\big[\big(\mathbf{H}^{\mathbf{b}}_{k,f}\big)^H\big(\hat{\mathbf{F}}^i_{k,f} + \mathbf{O}^{\mathbf{G},i}_{k,f}\big)\mathbf{H}^{\mathbf{b}}_{k,f}\big]. \tag{18}$$

The compensation matrix set $\boldsymbol{\Delta}^i = \big\{\mathbf{Z}^{\mathbf{A},i}_{k,f}, \mathbf{Z}^{\mathbf{C},i}_{k,f}, \mathbf{O}^{\mathbf{E},i}_{k,f}, \mathbf{O}^{\mathbf{F},i}_{k,f}, \mathbf{O}^{\mathbf{G},i}_{k,f}, \forall k, f\big\}$ is generated by the RL module detailed in section III-B2. All second-order expectations involved in equations (14)-(12) admit closed-form expressions, as detailed in our prior work [2].

The above operations enable direct precoder computation in each iteration using statistical information $\big\{\bar{\mathbf{H}}^{\mathbf{b}}_{k,f}, \Delta\mathbf{H}^{\mathbf{b}}_{k,f}, \forall k, f\big\}$, significantly accelerating convergence over SAA, particularly under severe channel inaccuracies. Moreover, note that both $\bar{\mathbf{H}}^{\mathbf{b}}_{k,f,n}$ and $\Delta\mathbf{H}^{\mathbf{b}}_{k,f,n}$ exhibit sparse structures with most energy concentrated in $B \ll M_t$ columns. By zeroing their negligible entries, RLDDU-Net reduces matrix multiplication complexity to $O\big(B^2\big)$, compared to $O\big(M_t^2\big)$ in WMMSE/WR-WMMSE. Similarly, the matrix $\tilde{\mathbf{B}}^i$ has most of its energy concentrated along the diagonal and in only $q < M_t$ rows/columns, allowing its inversion to be computed with complexity $O\big(q^3 + M_t\big)$, much lower than the $O\big(M_t^3\big)$ cost of computing $\big(\mathbf{B}^i\big)^{-1}$ in WMMSE/SWMMSE.

2) The RL Module: This module employ a stochastic policy network $\pi_{\boldsymbol{\theta}}$ to generate compensation matrices $\boldsymbol{\Delta}_{m,n} \triangleq \big\{\boldsymbol{\Delta}^1_{m,n}, \boldsymbol{\Delta}^2_{m,n}, \ldots, \boldsymbol{\Delta}^{I_{max}}_{m,n}\big\}$ and stopping coefficients $\boldsymbol{\beta}_{m,n} \triangleq$

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$\{\beta_{m,n}^1, \beta_{m,n}^2, \ldots, \beta_{m,n}^{I_{max}}\}$ for the DU module based on current CSI, with the network depth $I_{m,n} = \text{argmax}_i \, \beta_{m,n}^i$. Now, we discuss how to optimize $\pi_{\boldsymbol{\theta}}$. Although unsupervised learning using EWSR as the loss function is theoretically feasible, it involves a prohibitively long back-propagation path—from the DU network's output to the policy network's output, and then to the policy network's parameters—along with numerous matrix inversions, leading to high computational cost and significantly unstable gradients. To address this, we formulate the policy optimization problem as a contextual bandit (CB) and solve it via model-free RL [13].

Specifically, CB is a class of non-sequential decision-making problems [13]. At each step t of CB, a policy selects an action $\boldsymbol{a}_t \in \mathcal{A}$ according to the current state $\boldsymbol{s}_t \in \mathcal{S}$. The optimality of this action is reflected by an instantaneous reward signal $R(\boldsymbol{s}_t, \boldsymbol{a}_t) : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$, where the optimal action corresponds to $\max_{\boldsymbol{a}} R(\boldsymbol{s}_t, \boldsymbol{a}_t)$. In this work, we treat each downlink block as a single step in the CB setting. The n-th block in the m-th timeslot corresponds to the step $t = mN_b + n$. The state $\boldsymbol{s}_t \triangleq \{\bar{\mathbf{H}}_{k,f,m,n}^{\mathbf{b}}, \Delta\mathbf{H}_{k,f,m,n}^{\mathbf{b}}, \forall k, f\}$, for which the RL module sample an action $\boldsymbol{a}_t \triangleq \{\boldsymbol{\Delta}_{m,n}, \boldsymbol{\beta}_{m,n}\}$ from a Gaussian policy $\pi_{\boldsymbol{\theta}}(\boldsymbol{a}_t \mid \boldsymbol{s}_t)$. We parameterize the mean and variance of $\pi_{\boldsymbol{\theta}}(\boldsymbol{a}_t \mid \boldsymbol{s}_t)$ by neural networks $f_{\boldsymbol{\theta}^\mu}(\boldsymbol{s}_t)$ and $f_{\boldsymbol{\theta}^\delta}(\boldsymbol{s}_t)$, respectively, where $\boldsymbol{\theta} \triangleq \{\boldsymbol{\theta}^\mu, \boldsymbol{\theta}^\delta\}$. We define the reward function as $R(\boldsymbol{s}_t, \boldsymbol{a}_t) = \hat{\mathcal{R}}_{k,f,m,n} - \hat{\bar{\mathcal{R}}}_{k,f,m,n}$, where $\hat{\mathcal{R}}_{k,f,m,n}$ and $\hat{\bar{\mathcal{R}}}_{k,f,m,n}$ denote the EWSR performance of the $I_{m,n}$-layer RLDDU-Net and the I_{max}-layer DU network without compensation, respectively. Note that shallow depth $I_{m,n}$ may underfit due to limited model capacity, whereas an excessively large $I_{m,n}$ can result in overfitting due to the use of too many compensation matrices. The motivation behind this reward design is to encourage the adaptive-depth network with compensation matrices to achieve maximal performance improvement over the fixed-depth DU network.

Then, we solve the CB problem using model-free RL. Specifically, in the simulation, we adopt a stochastic successive convex approximation (SSCA)-based model-free RL algorithm from our prior work [14]. This algorithm estimates the policy gradient and constructs and solves a series of convex surrogate problems for the original policy optimization problem, ensuring that $\pi_{\boldsymbol{\theta}}$ ultimately converges to a stationary point of the CB problem. Note that the policy gradient estimation is based solely on immediate reward signals, without involving back-propagation through the DU network, which allows for more stable and simpler training. Moreover, in practical systems where parameters such as channel aging coefficients may drift over time, RL enables online adaptation, whereas unsupervised learning methods are typically limited to offline training.

Remark 1: Compared with existing robust precoding algorithms, RLDDU-Net offers three key advantages: 1) RLDDU-Net incorporates domain knowledge from the wideband SWMMSE algorithm, which is theoretically guaranteed to converge to a local optimum of problem $\mathbf{P}_1$, thereby achieving more stable performance across diverse scenarios than black-box networks in [8] and [9]; it employs approximation techniques that significantly accelerate convergence and reduce the computational burden of high-dimensional matrix operations and wideband multi-subcarrier processing, outperforming SAA-based and bi-level schemes in [3], [4], and [5]; and it leverages RL-driven design to further enhances its representational capacity over conventional DU schemes with fixed depth and compensation matrices, such as the PO-WMMSE Net in [2].

IV. Numerical Results

We consider a MU-MIMO-OFDM system where a BS with $M_t = 64$ antennas serves $K = \{5, 10, 15, 20\}$ users, each with $M_r = 2$ antennas. Channels are generated using the QuaDRiGa model [15] under the "3GPP_38.901_UMa_NLOS" scenario. Each timeslot lasts 1 ms and is divided in to $N_b + 1 = 7$ blocks. The carrier frequency is 3.5 GHz with a subcarrier spacing of 15 kHz. Users are randomly generated within a 200 m radius around the BS and move at 120 km/h in random directions. The total transmit power is $P = 1$ W. We set the receive noise power of all users to be equal, i.e., $\sigma_1^2 = \sigma_2^2 = \ldots = \sigma_K^2 = \sigma^2$, and adjust σ^2 such that $\text{SNR} = 10\log_{10}(1/\sigma_k^2)$ ranges from 0 dB to 20 dB. Since channel acquisition techniques are beyond the scope of this letter, we assume in the simulations, for simplicity, that the CSI acquired during the uplink training is accurate and that CSI errors at the BS are solely due to channel aging. Note that, however, the proposed scheme works for more practical case when the CSI errors are caused by both uplink training and channel aging. Following [2], we statistically obtain $\beta = \{0.96, 0.92, 0.84, 0.75, 0.63, 0.49\}$ corresponding to the six downlink blocks, where smaller α indicates more severe degradation. For RLLDU-Net, we set the number of sampled subcarriers frequency-domain interpolation to $\widetilde{F} = 8$. The policy networks $f_{\boldsymbol{\theta}^\mu}(\cdot)$ and $f_{\boldsymbol{\theta}^\delta}(\cdot)$ are composed of two convolutional layers (kernel size $H = 3$, channel number $C = 8$, and stride 1) and two fully connected layers (width $D = 128$). Baseline algorithms include wideband the SWMMSE [3], [4] and a wideband extension of PO-WMMSE Net [2] (i.e, perform the PO-WMMSE Net on the central subcarrier to obtain the precoder for the entire RBG), which are referred to as SWMMSE and PO-WMMSE Net for simplicity in the following. All baselines use 5 iterations/layers, while the RLDDU-Net adopts a maximum depth of $I_{max} = 5$. Additionally, since wideband SWMMSE can converges to a local optimum of problem $\mathbf{P}_1$ with sufficient iterations, we also simulate it with 100-iterations to estimate an upper performance bound.

Fig. 3a illustrates the EWSR performance of algorithms across various downlink blocks in a scenario with $K = 10$ users and $\text{SNR} = 20$ dB. The results show that SWMMSE, with sufficient iterations (i.e., the estimated upper bound), significantly outperforms WMMSE by accounting for CSI errors. However, with 5 iterations, SWMMSE provides only a marginal improvement over WMMSE due to slower convergence from the SAA method. In contrast, RLDDU-Net far exceeds WMMSE's performance and approaches the upper bound by directly leveraging CSI statistics. Simulation statistics reveal that RLDDU-Net requires an average depth of only $\{3.2, 3.2, 3.4, 3.7, 4.1, 4.6\}$ layers across the six

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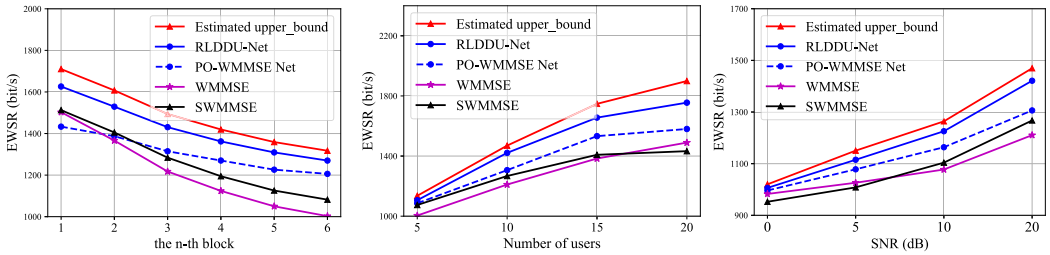

(a) (b) (c)

Fig. 3. (a) EWSR performance at different block. (b) EWSR performance versus different number of users. (c) EWSR performance versus different SNRs.

TABLE I
COMPLEXITY COMPARISON

Algorithms	Computational complexity	Values
(S)WMMSE	$O\left(M_t^2 M_r F K I_{max} + M_t^3 I_{max}\right)$	$2.31e7$
PO-WMMSE Net	$B^2 M_r^2 F K I + q^2 M_r K I + \left(M_t + q^3\right) I$	$2.85e6$
RLDDU-Net	$O\left(B^2 M_r^2 \tilde{F} K I + q^2 M_r K I + \left(M_t + q^3\right) I + C D M_t M_r \tilde{F} K\right)$	$1.26e6$

blocks, demonstrating superior convergence efficiency. Moreover, RLDDU-Net's adaptive compensation matrices further enhance performance compared to PO-WMMSE Net. While PO-WMMSE Net, with a fixed compensation matrix, outperforms WMMSE on average, it performs worse than WMMSE in some blocks. RLDDU-Net consistently outperforms all other baselines across all blocks, with performance gains becoming more significant in blocks with less accurate CSI. Fig. 3b shows the EWSR performance versus the number of users. As the number of users increases, the EWSR growth rate of all algorithms slows down due to higher interference and the amplified impact of inaccurate CSI. However, the performance gain of RLDDU-Net over the baselines increases, highlighting its superior interference management and robustness. Fig. 3c shows the performance at different SNRs, where RLDDU-Net consistently outperforms the baselines and approaches the estimated upper bound. Moreover, RLDDU-Net shows stronger performance at high SNR compared to the baselines, as performance at high SNR is primarily limited by interference, and RLDDU-Net is more effective at mitigating it.

Finally, we present the computational complexity and numerical results of RLDDU-Net and the baseline in Table I for the typical setup in Fig. 3a. The complexity of the former is relatively low because RLDDU-Net utilizes a low-complexity computational design by exploiting intermediate variables in the sparse structure of the beam domain and subcarrier correlations. The complexity of the latter also remains low due to the sparsity of the input to the policy network and the low dimensionality of the compensation matrices it outputs. As a result, it can be seen that the overall computational overhead of RLDDU-Net remains lower than that of PO-WMMSE and significantly lower than that of SWMMSE and WMMSE. This advantage becomes even more pronounced as M_t increases.

V. CONCLUSION

This letter designs RLDDU-Net for robust WMMSE precoding in massive MU-MIMO-OFDM systems. Derived from wideband SWMMSE, a robust precoding algorithm with theoretical guarantees, RLDDU-Net achieves strong performance under imperfect CSI. By introducing approximation techniques and leveraging beam-domain sparsity and subcarrier correlation, it greatly reduces computational complexity and accelerates convergence. Additionally, the RL-driven network in RLDDU-Net can adaptively generate compensation matrices to correct approximation errors and adjust network depth. Simulations demonstrate that RLDDU-Net achieves superior throughput with significantly reduced computational burden under imperfect CSI, offering a promising solution for practical WMMSE precoding deployment in real-world systems.

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